motion of microscopic particles becomes increasingly wild when they are examined and confined to ever smaller regions of space.

The uncertainty principle also gives rise to a striking effect known as quantum tunneling. If you fire a plastic pellet against a ten-foot-thick concrete wall, classical physics confirms what your instincts tell you will happen: The pellet will bounce back at you. The reason is that the pellet simply does not have enough energy to penetrate such a formidable obstacle. But at the level of fundamental particles, quantum mechanics shows unequivocally that the wave functions—that is, the probability waves—of the particles making up the pellet all have a tiny piece that spills out through the wall. This means that there is a small —but not zero—chance that the pellet actually can penetrate the wall and emerge on the other side. How can this be? The reason comes down, once again, to Heisenberg's uncertainty principle.

To see this, imagine that you are completely destitute and suddenly learn that a distant relative has passed on in a far-off land, leaving you a tremendous fortune to claim. The only problem is that you don't have the money to buy a plane ticket to get there. You explain the situation to your friends: if only they will allow you to surmount the barrier between you and your new fortune by temporarily lending you the money for a ticket, you can pay them back handsomely after your return. But no one has the money to lend. You remember, though, that an old friend of yours works for an airline and you implore him with the same request. Again, he cannot afford to lend you the money but he does offer a solution. The accounting system of the airline is such that if you wire the ticket payment within 24 hours of arrival at your destination, no one will ever know that it was not paid for prior to departure. In this way you are able to claim your inheritance.

The accounting procedures of quantum mechanics are quite similar. Just as Heisenberg showed that there is a trade-off between the precision of measurements of position and velocity, he also showed that there is a similar trade-off in the precision of energy measurements and how long one takes to do the measurement. Quantum mechanics asserts that you can't say that a particle has precisely such-and-such energy at precisely such-and-such moment in time. Ever increasing precision of energy measurements require ever longer durations to carry them out. Roughly speaking, this means that